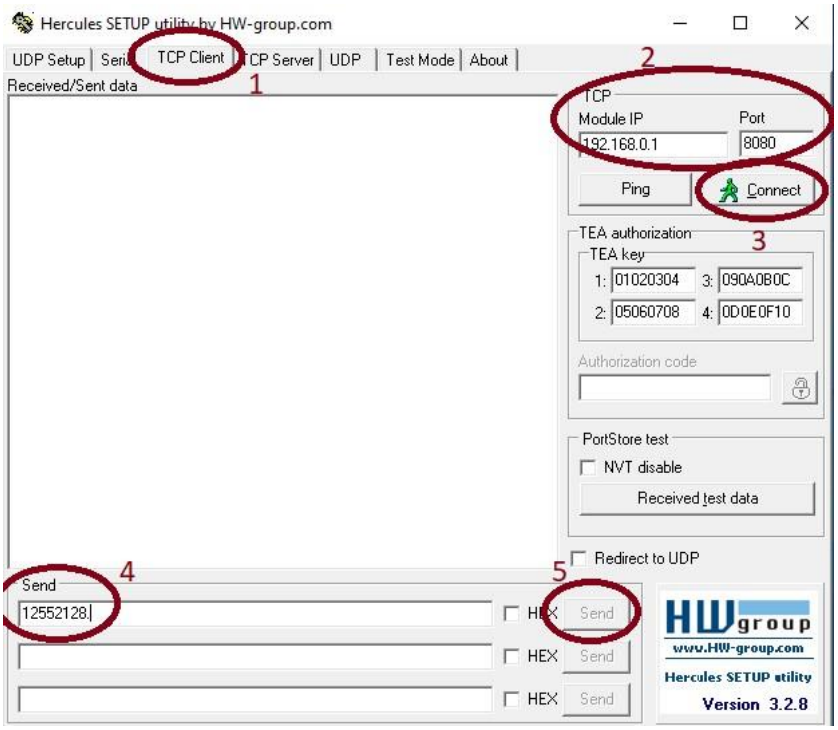
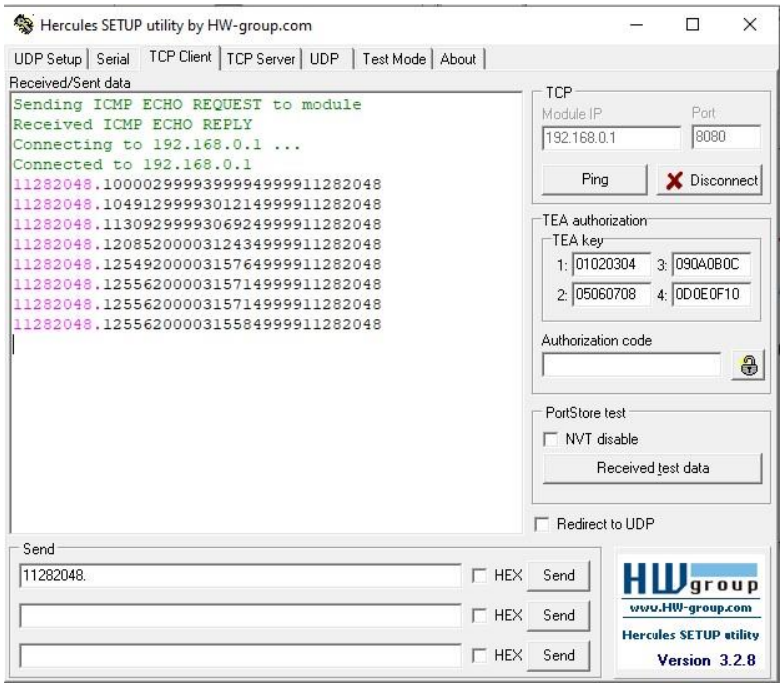


3. Once a request is sent, the WiFi DAQ will reply as shown below. The 4-20mA Inputs are mapped to 0 – 4095 and 4-20mA outputs are mapped to 0 – 255.



4. Use the formula for mapping: Min-max normalization: to [new_minA, new_maxA]

$$v' = \frac{v - \min_A}{\min_A \max_A - \min_A} (new_\max_A - new_\min_A) + new_$$

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5. Sample calculation for OP0 = 12mA, OP1 = 7mA: The data at (4) in the software might be **11282048.** (Please note: a dot (.) is used as a terminator). First 1 for Channel No. followed by 3digit value, followed by 2 for Channel No. and followed by 3-digit value.
   i. OP0 = (12-4)/(20-4)*(255-0)+0 = 127.5 ≈ 128
   ii. OP1 = (7-4)/(20-4)*(255-0)+0 = 47.8 ≈ 048
6. The reply from DAQ 11282048.12556200003155849999911282048 can be decoded based on Channel number followed by 4-digit value. The code received is separated as given below for understanding:



7. Leave the channel numbers (first digit) and use remaining 4-digit values as shown below:
   i. IP0 = (2556 - 0)/(4096-0)*(20-4)+4 = 13.984375 ≈ 13.99 mA
   ii. IP1 = (0 - 0)/(4096-0)*(20-4)+4 = 4 ≈ 4 mA
   iii. IP2 = (1558 - 0)/(4096-0)*(20-4)+4 = 10.0859375 ≈ 10.09 mA iv. IP3 = Open; A value of 9999 indicates open circuit

5.5 MQTT Communication

Node-Red, an IoT platform is used along with MQTT Explorer to perform MQTT communication in this experiment. Note the following points:

1. Refer the features of **MQTT Explorer** given in Section 3.6 and Node-Red given in Section 3.7.
2. Install Node-Red (If not installed). Refer the following link. https://nodered.org/docs/getting-started/windows
3. Install Modbus library using **Manage Palette** option.

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